

Generalized linear models

Logistic regression

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Logistic regression - Contents

Logistic regression

Logistic regression $1 - \pi(x) = 1 - \frac{\exp(-1)}{1 + \exp(-1)} = \frac{1}{1 + \exp(-1)}$

Equivalent representations: $\uparrow$

1.

$$E[Y|X]$$

$$\pi(\mathbf{x}) = \frac{\exp(\mathbf{x}^\top \beta)}{1 + \exp(\mathbf{x}^\top \beta)}$$

" $\pi(x) = P(Y=1|x) \rightarrow$ shows effect of x on prob."

2.

$$\frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})} = \exp(\mathbf{x}^\top \beta)$$

"Chance", "Odds" $\Rightarrow \frac{\pi(x)}{1 - \pi(x)} = \exp(\beta_0 + \beta_1 x)$
 $= \exp(\beta_0) \cdot \exp(\beta_1 x)$
 $\rightarrow$ multiplicative

3.

$$\log\left(\frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})}\right) = \mathbf{x}^\top \beta$$

"logits", "log-odds", "log-chances"; determined linearly

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1. For one-dimensional predictor

$$\pi(x) = F(\beta_0 + \beta_1 x_1) \quad \text{with} \quad F(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)}$$

→ $F(\eta)$ is a distribution function, i.e.:

- lies in between 0 and 1 with $\lim_{\eta \rightarrow -\infty} F(\eta) = 0$
and $\lim_{\eta \rightarrow \infty} F(\eta) = 1$

- strictly monotonously increasing, 1st derivative is always positive

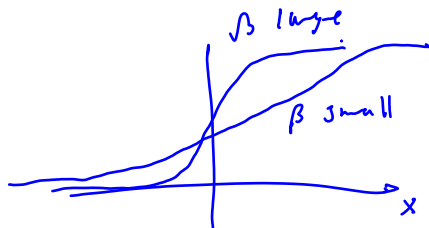
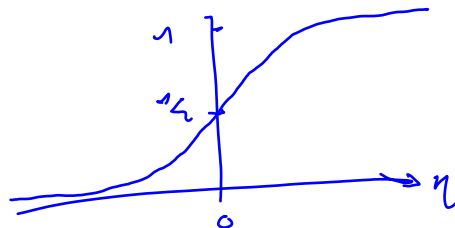
$$F'(\eta) = \frac{\exp(\eta)(1 + \exp(\eta)) - \exp(\eta)^2}{(1 + \exp(\eta))^2} = \frac{\exp(\eta)}{(1 + \exp(\eta))^2} > 0$$

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Further characteristic: $F(\cdot)$ is point-symmetric:

► $F(0) = 0.5$ and $F(\eta) = 1 - F(-\eta)$

→ for $x_1 = 0$, $F(\cdot)$ is only depending on β_0
→ β determines the slope / „steepness“



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Examples

(1)

- ▶ y_i : car in household
- ▶ x_i : net income (in €)
- ▶ β_0 : -1.86
- ▶ β_1 : 0.002

(2)

- ▶ y_i : unemployment (short / long)
- ▶ x_i : age
- ▶ β_0 : 2.235
- ▶ β_1 : -0.0347



⇒ interpretation of the β 's not easy

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Interpretation for β_1

(1) Sign

- ▶ $\beta_1 < 0 \Rightarrow \pi(x_1) \searrow$ when $x_1 \nearrow$
- ▶ $\beta_1 = 0 \Rightarrow$ no influence of x_1 on $\pi(x_1)$
- ▶ $\beta_1 > 0 \Rightarrow \pi(x_1) \nearrow$ when $x_1 \nearrow$

$$\pi(x_1) = \frac{\exp(\eta)}{1 + \exp(\eta)} = \frac{1}{1 + \exp(-\eta)} = \frac{1}{1 + \exp(-\beta_0 - \beta_1 x_1)}$$

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(2) Approximation

Assuming $\pi(x_1) = 0.5$ and that x_1 is modified by $\frac{1}{\beta_1}$ to $\tilde{x}_1 = x_1 + \frac{1}{\beta_1}$.

Then:
$$\frac{\pi(\tilde{x}_1)}{1 - \pi(\tilde{x}_1)} = \exp(\beta_0 + \beta_1 \tilde{x}_1) = \exp(\beta_0 + \beta_1 x_1 + 1)$$
$$= \frac{\pi(x_1)}{1 - \pi(x_1)} \exp(1) = \frac{0.5}{1 - 0.5} \cdot e = e$$

$$\Rightarrow \pi(\tilde{x}_1) = e (1 - \pi(\tilde{x}_1)) \quad (*) \quad \frac{e}{1 + e} \approx 0.73$$

If $\pi(x_1) = 0.5$ and change $\tilde{x}_1 = x_1 + \frac{3}{\beta_1}$

with analogous
calculations

$$\pi(\tilde{x}_1) = \frac{e^3}{1 + e^3} \approx 0.95$$

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reference
and if net income x_1 is such that
 $\pi(x_1) = 0.5$

In the examples

(1) Car purchase

net income increases by $\frac{1}{0.002} = 500 \text{ €}$

$\Rightarrow$ Prob. increases from 0.5 to 0.73

net income increases by $\frac{3}{0.002} = 1500 \text{ €} \Rightarrow$ prob.

(2) Unemployment increases from 0.5 to 0.95

age decreases by $\frac{1}{-0.0447} = -22.37$

$\Rightarrow$ Prob. for short term unemployment increases from 0.5 to 0.73

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- (3) The odds in (2.) from Slide 3 change by the increase of x_1 to $\tilde{x}_1 = x_1 + 1$ as follows:

$$\frac{\pi(\tilde{x}_1)}{1 - \pi(\tilde{x}_1)} = \exp(\beta_0 + \beta_1(x_1 + 1)) = \frac{\pi(x_1)}{1 - \pi(x_1)} \cdot \exp(\beta_1)$$

$\Rightarrow$ multiplicative change by the factor $\exp(\beta_1)$

$\Rightarrow \exp(\beta_1)$ acts as a factor on the odds/chances, when x_1 is changed to $x_1 + 1$

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In the examples

(1) Car purchase $\exp(0.002) = 1.002 > 1$

$\Rightarrow$ every € increases the chances/odds by 1.002
($\hat{=}$ 0.2 %) independent from the original x value

(2) Unemployment

$$\exp(\beta_1) = \exp(-0.0447) = 0.96 < 1$$

$\Rightarrow$ every year of age decreases the chances
for short term unempl.

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Centering and rescaling of covariates

- ▶ Sometimes it's better to use a different scaling for variable x and/or to center it

- ▶ Car purchase example: Regard income per 1000 € & to center it

$$\ln\left(\frac{\pi}{1-\pi}\right) = \beta_0 + (\text{income per 1000€} - 3) \cdot \beta_1$$

income $\rightarrow$ centering

$$\Rightarrow \beta_0 = 1.35 \Rightarrow \text{linear predictor for average hh}$$
$$\Rightarrow \frac{\exp(1.35)}{(1 + \exp(1.35))} = 0.79$$

$$\Rightarrow \beta_1 = 2.14 \Rightarrow \exp(2.14) = 8.5$$

$\Rightarrow$ Factor by which the odds are increased if the income changes by one unit ($\hat{=}$ 1000 €)